Unit 1 – Using Objects and Methods
HW 3

Name: **KEY**

1) Trace the following code segments to determine what is printed when they run.

a)
```
int x = 5;
int y = 10;
int z = y % 3 + x - 9;
System.out.println(z);
```

−3

b)
```
double a = 2.5;
double b = 0.25;
double c = 3.0;
double d = a + b + c / 4;
System.out.println(d);
```

3.5

c)
```
int value = 11 % 4;
int count = 5 / 3;
System.out.println(value + count);
```

4

d)
```
double answer = 3 * 4 / 2 % 5;
double result = (2 + 2 - answer) / 2;
System.out.println(result);
```

1.5

2) What would the following expressions evaluate to?

```
int x = 12;
```

x++	x--	x += 5	x -= 5	x *= 5	x /= 5	x = x + x / x
13	11	17	7	60	2	13

3) Trace the following code segments.

a) What is the value of z?

```
int x = 5;
int y = 10;
int z = ++x * ++y;
```

++(5) * ++(10)

6 * 11

66

b) What is the value of result?

```
int a = 10;
int b = 5;
int c = 1;
int result = a + c - ++b;
```

10 + 1 – ++(5)

10 + 1 – 6

11 – 6

5

4) On the left, there are code segments that are incorrect. On the right, rewrite the code segments so that they will work.

Incorrect Code Segment	*Correct Code Segment*

`int a;`
`int b;`
`System.out.println(a + b);`
	`int a = 4;`
	`int b = 4;`
	`System.out.println(a + b);`

`int c;`
`int d = c;`
`System.out.println(d);`
	`int c = 4;`
	`int d = c;`
	`System.out.println(d);`

`e = 5;`
`f = e + 3;`
`System.out.println(f);`
	`int e = 5;`
	`int f = e + 3;`
	`System.out.println(f);`

`int g = 3;`
`int h;`
`g + 4 = h;`
`System.out.println(h);`
	`int g = 3;`
	`int h;`
	`h = g + 4;`
	`System.out.println(h);`

`j = k = 3;` `System.out.println(j + k);`	`int j = 3, k = 3;` `System.out.println(j + k);`

5) What is the difference between `System.out.print()` and `System.out.println()`?

`print()` prints the text on the same line without moving to the next line. `println()` prints the text and then moves the cursor to the next line, so the next output appears on a new line.

6) Explain why `System.out.println(10 / 4);` prints 2 instead of 2.5.

Since 10 and 4 are both integers, Java performs integer division and ignores the decimal part, so 10 / 4 gives 2 instead of 2.5.

7) What happens if a user enters a string instead of an integer when using `keyboard.nextInt()`?

If the user enters a string instead of an integer, the program will crash (this is called a run-time error) because `nextInt()` expects an integer but receives a string.